Week 7 Assignment: Module Seven Activity – Part Two

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Introduction

Public transportation is a critical equity issue, and understanding how it is used across different service areas is essential for any nonprofit working to improve access. This report analyzes fiscal year 2023 ridership data from the National Transit Database for three Washington-state transit agencies that represent the "neighborhoods" identified in the scenario: King County, Spokane Transit Authority, and the City of Longview. Three Python-generated visualizations form the basis of the analysis, exploring total ridership, the modal composition of each system, and per-capita utilization across the three communities.

Trends, Differences, and Outliers

The most striking trend in the data is the sheer scale of difference between the three neighborhoods. King County recorded approximately 78.9 million unlinked passenger trips in FY 2023, while Spokane Transit Authority reported about 9.4 million trips and the City of Longview reported only 274,494. King County's ridership is roughly 287 times that of Longview and more than eight times that of Spokane. The City of Longview is clearly an outlier in absolute terms; however, its service area population is also significantly smaller than the other two, so absolute comparisons alone can be misleading. King County is also unique in operating a multimodal system that includes bus, ferry, and rail service, while Spokane and Longview rely entirely on bus service.

Disparities and Patterns Relevant to the Nonprofit

When ridership is normalized against service-area population, a more meaningful disparity emerges. King County residents take an average of 33.6 transit trips per year, Spokane residents take 20.0, and Longview residents take only 4.5. This per-capita metric removes the population-size effect and reveals how intensively each community actually uses its transit system. The roughly sevenfold gap between King County and Longview is far more relevant to the nonprofit's mission than the raw 287-fold gap in absolute totals, because it points to potential equity issues in service quality, frequency, or coverage rather than simply reflecting that Longview is a smaller place. Spokane's middle position—about two-thirds the per-capita rate of King County—suggests there is room for utilization growth even in a moderately sized system

Summary of Transit Usage Across Neighborhoods

Taken together, the data suggests that transit usage across the three neighborhoods is not just unequal in scale, but unequal in quality of access. King County operates the most diversified and most heavily used system, Spokane operates a moderately productive bus-only network, and Longview operates a small bus system with very low per-capita ridership. The pattern is consistent across all three visualizations. The data does not explain why Longview ridership is so low, but it clearly identifies Longview as the service area in greatest need of further investigation and possible intervention.

How the Visualizations Support These Conclusions

The three visualizations work together to build this conclusion. The bar chart of total annual ridership establishes the most basic comparison and immediately exposes Longview as an outlier. The grouped bar chart of ridership by transit mode adds context by showing that King County's lead is driven primarily by bus service, but its ferry and rail offerings represent forms of mobility that are simply unavailable to residents of the other two neighborhoods. Finally, the per-capita ridership chart corrects for the population-size confound that makes the first chart misleading on its own. Without this third chart, a reader might wrongly conclude that Longview's transit problem is just a matter of being a small city. Together, the three views move the analysis from descriptive—how much—to comparative—how much relative to population—to diagnostic—where the nonprofit should focus.

Strategic Implications for the Nonprofit

For the nonprofit, the data points toward a clear set of strategic priorities. Longview, with the lowest per-capita ridership and a bus-only system, should be the priority for further research into barriers such as service frequency, route coverage, accessibility, fare structure, or community awareness. Spokane, while performing better, has room to grow utilization through targeted improvements rather than full system expansion. King County serves as a useful benchmark for what diversified, well-utilized transit can look like, but its scale should not be treated as the goal for smaller agencies. Effective advocacy will involve setting realistic, population-adjusted goals for each service area rather than comparing them on absolute volume alone.

The Importance of Data-Driven Insights

This analysis demonstrates why data-driven decision making matters when addressing real-world problems. Without normalizing for population, a well-intentioned nonprofit might have focused its resources on King County simply because the absolute numbers are largest, when in fact the most pressing equity question lies in Longview. Visualizations make these distinctions immediately visible to stakeholders who may not work with raw spreadsheets every day. In a sector where funding is limited and advocacy attention is finite, allowing data to guide where the nonprofit invests its energy ensures that effort is directed where it is most likely to benefit the communities being served.

AI Usage Acknowledgment

In accordance with the assignment guidelines, generative AI tools (Claude by Anthropic) were used to support data exploration, code structuring for the Jupyter Notebook, revision and editing of this report. All analytical conclusions were reviewed and verified against the underlying dataset. The dataset itself is the February 2025 release of the National Transit Database (NTD) Complete Monthly Ridership file published by the U.S. Federal Transit Administration.